

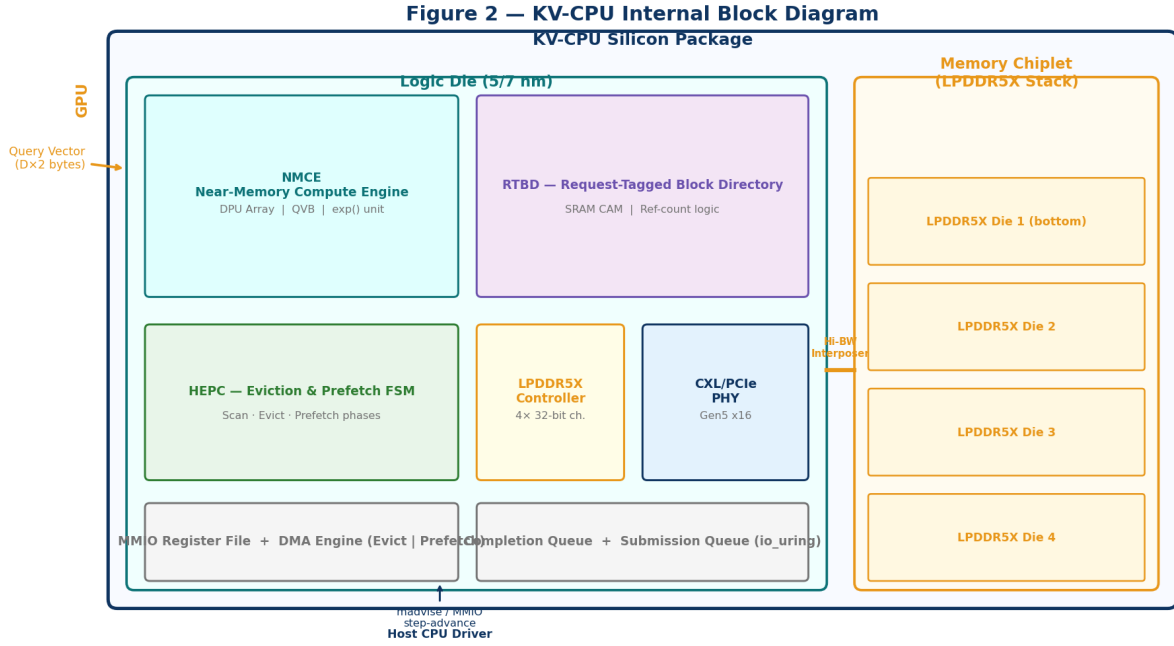


Figure 2 — KV-CPU Internal Block Diagram: Logic Die (NMCE, HEPC, RTBD, CXL PHY) and Memory Chiplet (LPDDR5X Stack)

5.2 Pillar I — Near-Memory Compute Engine (NMCE)

5.2.1 Why NMCE is Not Generic PIM

Processing-In-Memory architectures such as Samsung HBM-PIM implement general-purpose GEMV for weight access. The NMCE is categorically different in three respects. **First**, its operation is semantically specific to scaled dot-product attention — inputs are a query vector Q_h and a KV block $K[0..B-1]$, a data structure existing only in transformer inference. **Second**, the output (B attention score scalars) is not a final computation result but a signal that drives the HEPC priority update for the block just scored. A generic PIM unit produces a result and stops; the NMCE produces a result that changes the hardware eviction policy for the very memory it accessed. **Third**, the NMCE can only function correctly if the RTBD has identified that the required block is resident in T1 and provided its physical address — meaningless without the RTBD's request-tagged block tracking. The NMCE is an integral node in the closed-loop KV lifecycle system, not a separable compute unit.

5.2.2 Operation and PCIe Traffic Reduction

When the GPU identifies a KV block needed for attention is resident in T1, it issues an NMCE Compute Descriptor to the KV-CPU submission queue (CXL.io BAR) containing: query vector Q_h ($D \times 2$ bytes, FP16/BF16), T1 address range of key block $K[0..B-1]$, and scaling factor $1/\sqrt{D}$. The NMCE executes entirely on-device:

1. Fetches $K[0..B-1]$ from T1 LPDDR5X into local key line buffer — zero PCIe traffic
2. Computes $\text{score}[i] = Q_h \cdot K[i] \times (1/\sqrt{D})$ for i in $[0, B-1]$ using D -wide DPU array
3. Applies piecewise-linear $\exp()$ approximation (8 segments, $\pm 0.1\%$ relative error)
4. Transmits B scalar scores ($B \times 2 = 64$ bytes for $B=32$, FP16) to GPU via CXL.io peer-to-peer

Traffic quantification ($D=128$, $B=32$, FP16): Without NMCE: 8,192 bytes per block. With NMCE: 320 bytes. Ratio = 25.6x. With 64-head QVB amortisation: approaching 128x.

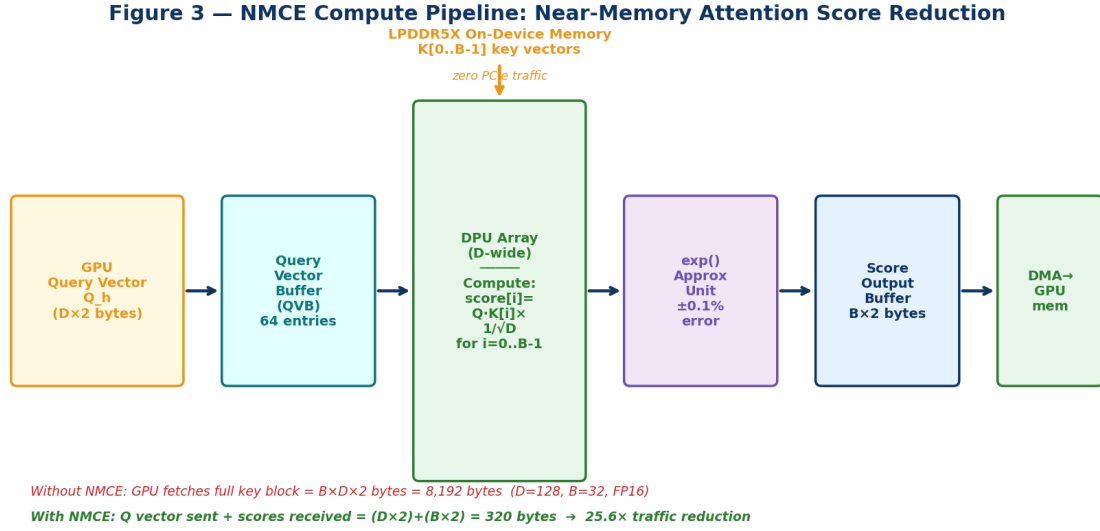


Figure 3 — NMCE Compute Pipeline: Attention-Specific Near-Memory Score Reduction (zero PCIe traffic for key vectors)

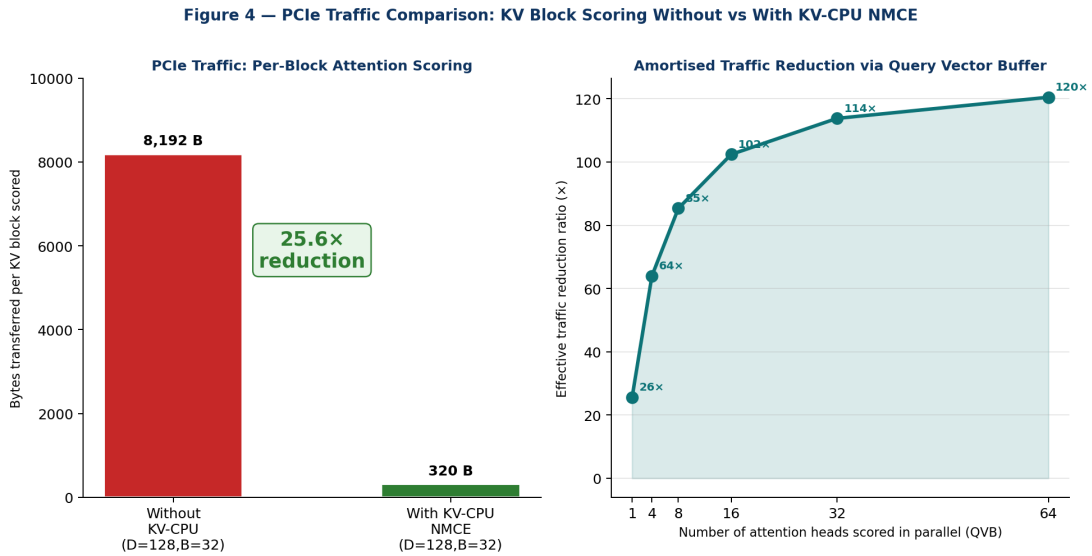


Figure 4 — PCIe Traffic Comparison: Per-block scoring without vs. with NMCE; amortised multi-head reduction via QVB

5.2.3 NMCE Hardware Components

- D-wide Dot-Product Unit (DPU) array: configurable $D=64/128/256$, FP16/BF16 MAC
- Query Vector Buffer (QVB): 64-entry $\times$ D-element SRAM — amortises query transmission across up to 64 attention heads
- Softmax $\exp()$ approximation unit: 8-segment piecewise linear, $\pm 0.1\%$ relative error
- Key Line Buffer: $2 \times B \times D \times 2$ byte ping-pong SRAM for double-buffered key prefetch from T1
- Completion DMA Engine: writes $B \times 2$ byte score vector to GPU-mapped BAR via CXL.io peer-to-peer write

5.3 Pillar II — Hardware Eviction and Prefetch Controller (HEPC)

5.3.1 Why Software Cannot Solve the Decode-Step Awareness Problem

Standard eviction policies (LRU, LFU, ARC, CLOCK) rank blocks by access history. For transformer KV-cache, access history is insufficient because the access pattern is structurally determined by the decode step. As decoding progresses, attention patterns change in model-specific ways: system prompt blocks are accessed almost every step; recent-context blocks frequently; distant-context blocks may be dormant for hundreds of steps then re-accessed when attention shifts. A policy without the current decode step number cannot distinguish "block is unimportant" from "block is temporarily outside the attention window." The HEPC's step-advance register receives the current decode step from the GPU, making the hardware policy semantically aware of the inference session's temporal position — impossible for any software-only approach.

Figure 5 — HEPC Three-Phase State Machine

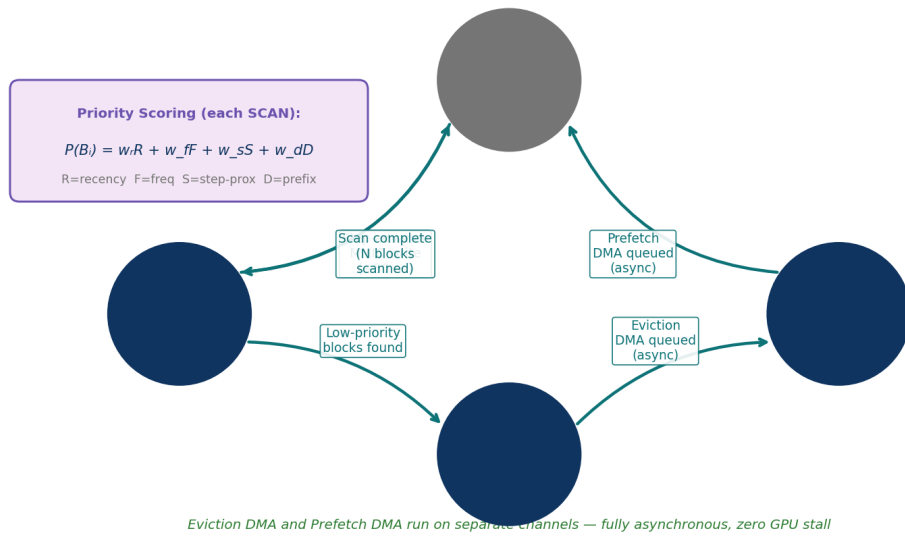


Figure 5 — HEPC Three-Phase State Machine: Scan/Evict/Prefetch triggered by GPU MMIO step-advance signal (~50 ns)

5.3.2 Priority Scoring Function

$$P(B_i) = w_r \times R(B_i) + w_f \times F(B_i) + w_s \times S(B_i) + w_d \times D(B_i) \pmod{2^{16}}$$

Symbol	Component	Hardware Implementation	Why Transformer-Specific
R(B _i)	Recency	Saturating 8-bit counter reset to 255 on access; decremented by 1 per step-advance signal. Range 0–255.	Captures temporal locality in decode steps, not wall-clock time. Decay rate = decode steps.
F(B _i)	Frequency	Saturating 8-bit counter incremented on each block access. Never decremented. Range 0–255.	Captures working set membership — system prompt blocks hit every step saturate to 255 rapidly.
S(B _i)	Step proximity	max(0, W - (t _{current} - t _{last_access})). W = programmable window register.	Key novelty: measures recency in decode steps. A block unaccessed for W steps predicted outside attention window. No generic PIM equivalent.
D(B _i)	Prefix dependency	Binary flag from RTBD is_prefix_block field. Weighted by w _d (default 200, max weight).	Shared prefix blocks must never be evicted while any request references them. Hardware reference count enforces this at silicon speed.

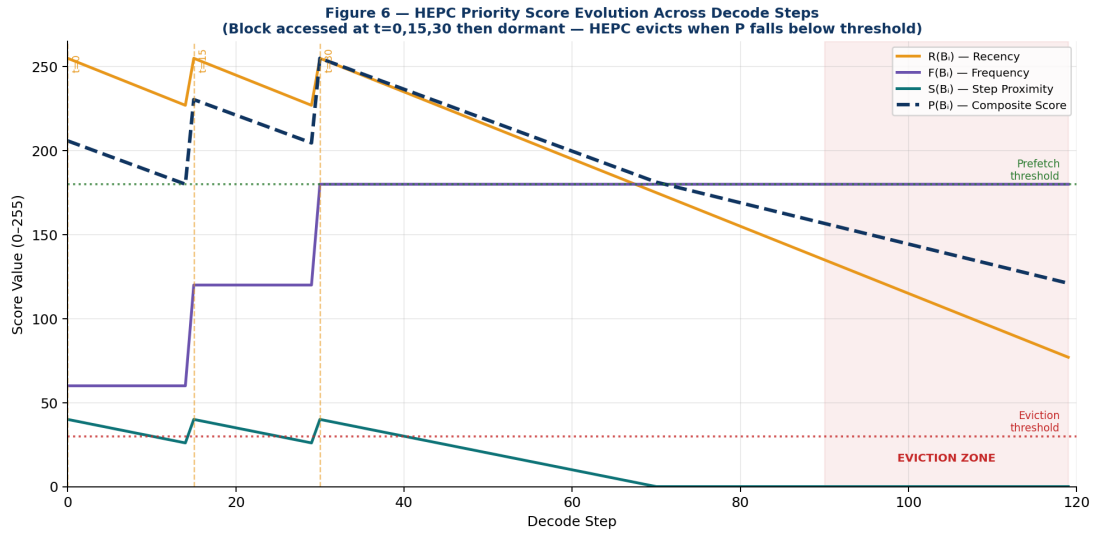


Figure 6 — Priority Score Evolution: Block accessed at t=0,15,30 then dormant; HEPC evicts when P falls below threshold

5.3.3 Three-Phase HEPC Cycle

1. **SCAN PHASE:** Triggered by GPU MMIO write to step-advance register (~50 ns). HEPC scans all active RTBD entries in parallel using pipelined comparator array, recomputing $P(B_i)$ per block in one clock cycle each.
2. **EVICT PHASE:** Blocks with $P(B_i) < \text{eviction_threshold}$ placed in eviction DMA queue. Background DMA transfers to T2 (CXL.mem write) or T3 (io_uring opcode). Separate DMA channel from PREFETCH — no serialisation.
3. **PREFETCH PHASE:** Blocks in T2/T3 with $P(B_i) > \text{prefetch_threshold}$ staged into T1 via background DMA. Triggered proactively before GPU requests — GPU never waits on a correctly prefetched block.

5.4 Pillar III — Request-Tagged Block Directory (RTBD)

Figure 7 — RTBD Entry Bit-Field Layout (Total: ~240 bits / 30 bytes per entry)

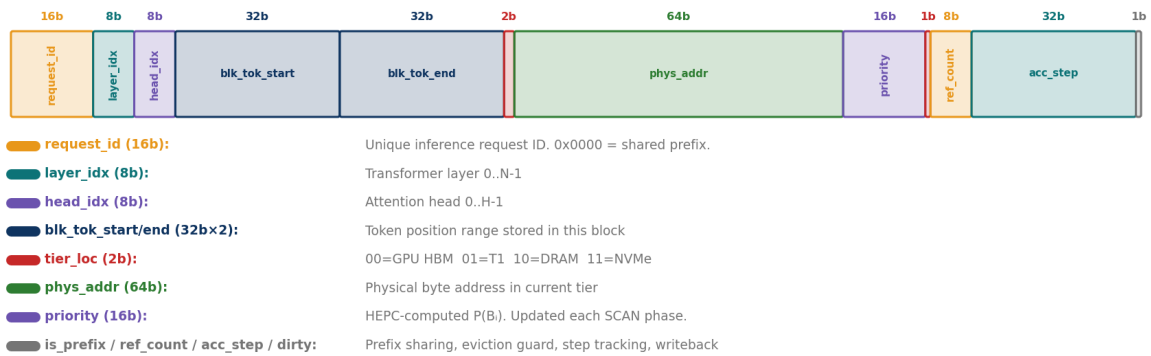


Figure 7 — RTBD Entry Bit-Field Layout: ~240 bits (30 bytes) per entry; up to 65,536 entries in SRAM CAM

5.4.1 Structure and Single-Cycle Lookup

The RTBD is a fully-associative Content-Addressable Memory (CAM) in SRAM on the KV-CPU logic die. Each entry associates a physical KV block with: request_id (16b), layer_idx (8b), head_idx (8b), block_token_start

(32b), block_token_end (32b), tier_location (2b: 00=GPU HBM, 01=T1, 10=T2, 11=T3), physical_addr (64b), priority_score (16b), is_prefix_block (1b), reference_count (8b), access_step (32b), dirty_bit (1b). The CAM provides single-cycle lookup given a (request_id, layer_idx, head_idx, block_token_start) tuple.

5.4.2 Hardware Prefix Block Sharing via Reference Counting

Figure 8 — RTBD Prefix Block Sharing: Hardware Reference Counting
(Prefix length=4096 tokens, 70B model FP16 → 10.7 GB per prefix copy)

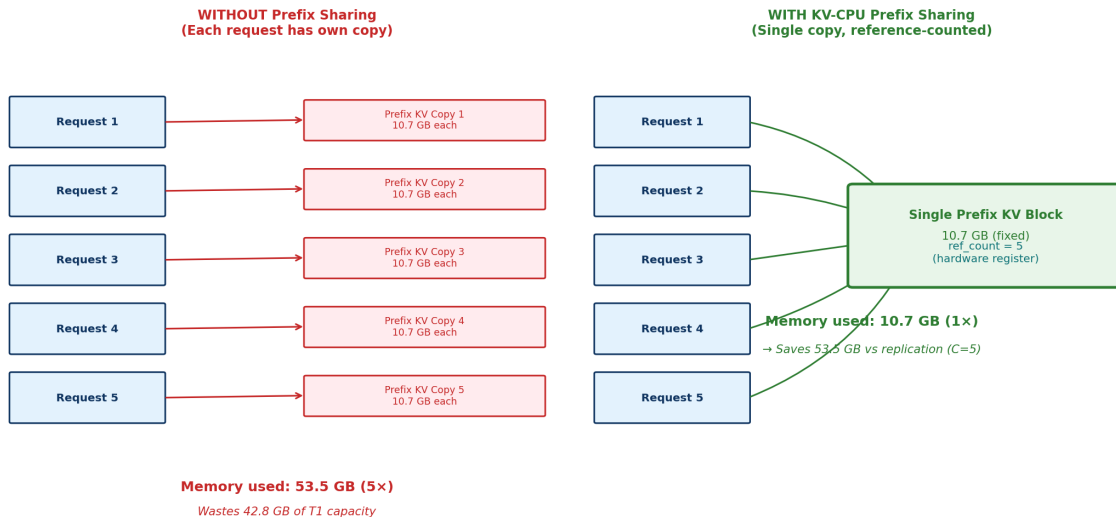


Figure 8 — Prefix Block Sharing: Hardware reference counting vs. C-fold replication (10 requests, 70B, 4096-token prefix = 10.7 GB)

1. Prefix blocks registered with `is_prefix_block=1` and `request_id=0x0000` stored as single physical copy
2. Additional requests issue `RTBD_SHARE` (CXL.io register write), incrementing hardware `reference_count`
3. NMCE scores prefix blocks using single physical copy — zero data duplication
4. `is_prefix_block=1` AND `reference_count > 0` = hardware eviction guard; HEPC cannot evict regardless of priority
5. On request termination, `RTBD_RELEASE` decrements `reference_count`. Eviction eligible only when `reference_count = 0`

5.5 Pillar IV — Kernel Driver as Hardware Control Plane

5.5.1 Why Pillar IV is a Hardware Invention, Not a Software Invention

Patent offices sometimes challenge kernel driver claims as software inventions. The `kv_cpu.ko` driver is correctly characterised as a **hardware control plane** because every user-visible interface terminates in a write to a hardware register in the KV-CPU silicon. The HEPC priority scoring function cannot execute correctly without receiving the decode step via the step-advance register write. The RTBD eviction guard for prefix blocks cannot activate without `RTBD_SHARE/RTBD_RELEASE` register commands. The NMCE cannot locate blocks without RTBD CAM lookup. **Remove the driver: the hardware policy loop cannot close. Remove the hardware: the driver is meaningless.** They are co-dependent components of a single hardware-software system. The software interface exists solely to convey LLM-semantic information (decode step, block hotness, prefix membership) to hardware that has no other path to receive it.

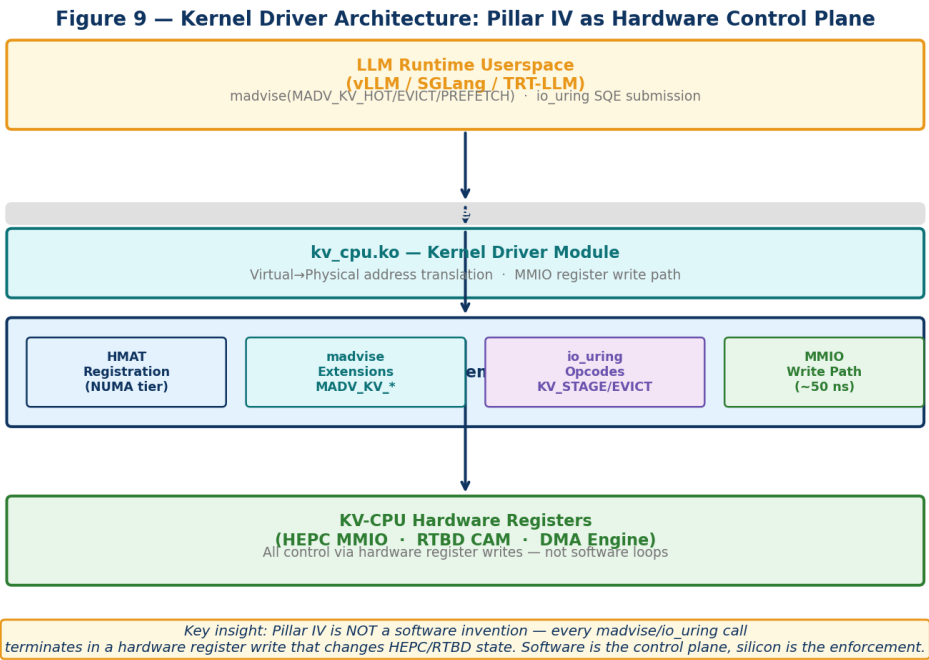


Figure 9 — Kernel Driver as Hardware Control Plane: Every API call terminates in a hardware register write

5.5.2 HMAT Memory Tier Registration

The driver registers T1 LPDDR5X with the Linux kernel HMAT subsystem: read latency 150 ns local / 220 ns remote NUMA; write latency 155/225 ns; peak read bandwidth 200 GB/s; NUMA node ID assigned dynamically. This causes the kernel memory management subsystem to treat T1 as a first-class NUMA node addressable via `mbind()`, `set_mempolicy()`, and `numactl`.

5.5.3 Semantic Memory Hint Interface: Three `madvise` Extensions

Hint	Arguments	Hardware Register Written	Effect on HEPC/RTBD State
MADV_KV_HOT	addr, length	HEPC priority boost register (MMIO write)	All matching RTBD entries: <code>priority_score</code> set to 0xFFFF. Block excluded from eviction during active decode loop.
MADV_KV_EVICT	addr, length	HEPC immediate-evict register (MMIO write)	Matching blocks placed in eviction DMA queue immediately, bypassing normal scan phase.
MADV_KV_PREFETCH	addr, length, step, lookahead	HEPC step register + W window register (MMIO writes)	<code>S(B_i)</code> recomputed; blocks with <code>S > prefetch_threshold</code> immediately scheduled for prefetch DMA.

5.5.4 `io_uring` Integration for Zero-Copy Tier Migration

- IORING_OP_KV_STAGE:** Async migration of KV block from T2/T3 to T1. Uses kernel-registered fixed buffers for zero-copy DMA. CQE signals GPU runtime that block is staged and RTBD is updated.
- IORING_OP_KV_EVICT:** Async migration from T1 to T2 or T3. CQE triggers atomic update of `tier_location` and `physical_addr` fields in RTBD for the migrated block.

5.6 Minimal Embodiments — Three Independent Operating Modes

Each pillar is independently operable. Three minimal embodiments exist in addition to the full four-pillar device, ensuring each independent claim survives even if other claims face rejection:

Figure 11 — Minimal Embodiments: Each Pillar is Independently Operable
(Embodiment A = Claim 1 only · B = Claims 5+9 only · C = All claims active)

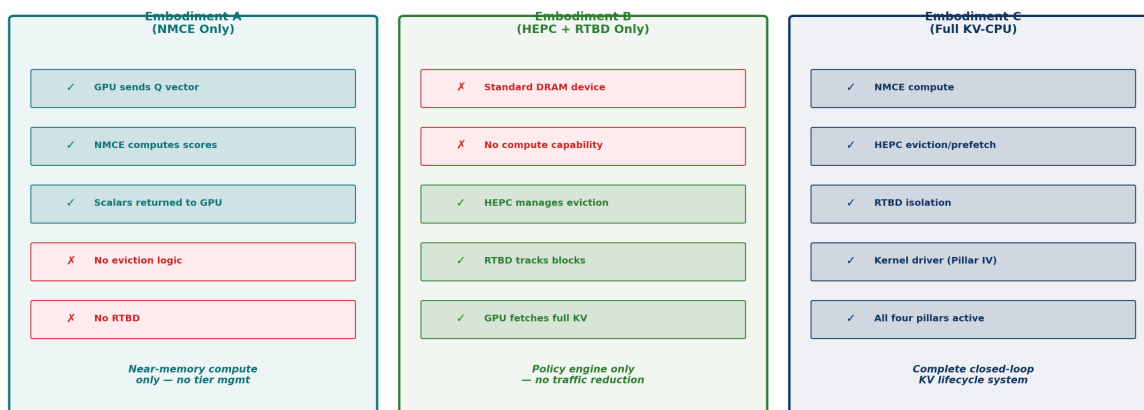


Figure 11 — Minimal Embodiments: A=NMCE only (Claims 1–4), B=HEPC+RTBD only (Claims 5–12), C=Full device (all claims)

Embodiment A — NMCE-Only Device (Pillar I alone)

Device comprising only NMCE and LPDDR5X memory, without HEPC or RTBD. Host LLM runtime manually stages KV blocks into T1 and issues NMCE compute descriptors. Still delivers 25.6x PCIe traffic reduction. Claims 1–4 read on this embodiment.

Embodiment B — HEPC + RTBD Only (Pillars II + III, no compute)

Device comprising HEPC, RTBD, and T1 LPDDR5X, without NMCE. Functions as intelligent KV-cache memory tier with hardware eviction and request isolation. GPU fetches full KV vectors across PCIe, but eviction latency drops from 200–600 μ s to ~50 ns and prefix sharing is provided. Claims 5–12 read on this embodiment.

Embodiment C — Full Four-Pillar KV-CPU (preferred embodiment)

Complete device combining all four pillars in closed-loop configuration. Subject of all claims.

5.7 Hardware Implementation

- **Logic Die:** NMCE, HEPC FSM, RTBD SRAM CAM, LPDDR5X PHY, CXL/PCIe 5.0 PHY, MMIO register file, dual DMA engine (evict / prefetch channels). Process: TSMC N5 or N7. Estimated die area: 40–80 mm².
- **Memory Chiplet:** 2–4 LPDDR5X die stacks (128–512 GB capacity). Connected via silicon or organic interposer at 55 μ m pitch. 4x 32-bit channels at 8.5 Gbps/pin = 256 GB/s aggregate.
- **Power:** Active ~20–28 W; Idle ~8–12 W; Standby ~3 W. Form factor: PCIe HHHL or CXL EDSFF E3.S.
- **Platform compatibility:** Intel Xeon Scalable (Sapphire/Granite Rapids, CXL 1.1/2.0); AMD EPYC (Genoa/Turin, CXL 1.1); ARM Neoverse N2/V2; AWS Graviton 3/4; Ampere Altra Max (PCIe 5.0). Linux kernel 6.2+.

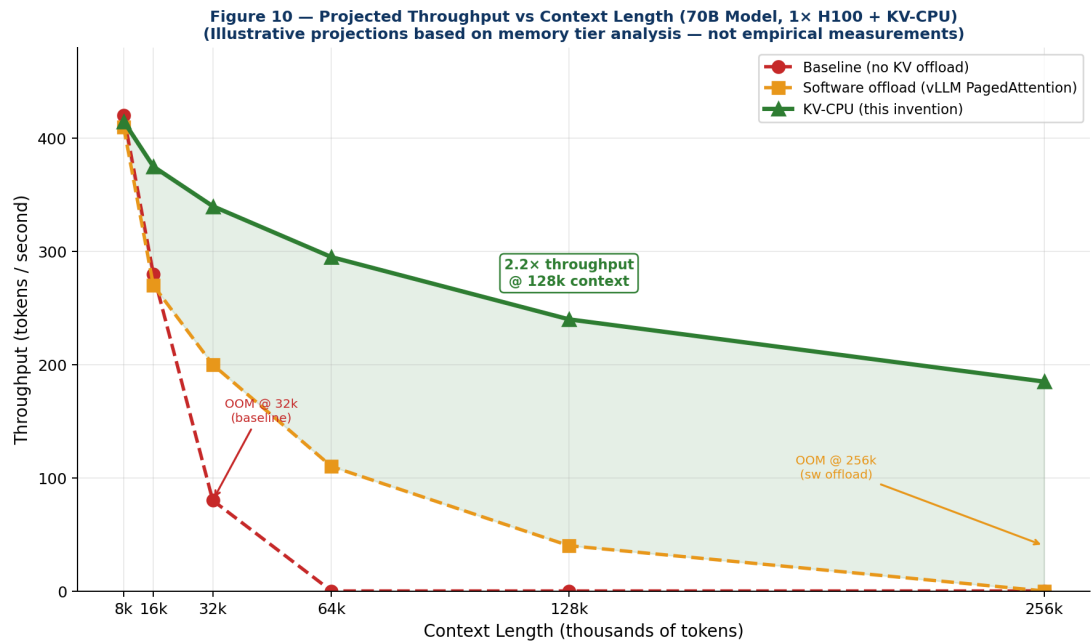


Figure 10 — Projected Throughput vs Context Length: 70B model, 1x H100 + KV-CPU (illustrative projections)

Metric	Baseline (no KV-CPU)	With KV-CPU	Improvement
Max context length (70B, 1x H100)	~20k tokens (HBM limit)	~256k tokens (T1 limit)	~12.8x context expansion
PCIe traffic per KV block scored	8,192 bytes	320 bytes	25.6x reduction (128x with QVB)
KV block score latency (T1)	200–600 μs (DMA + SW)	~0.25 μs (NMCE on-device)	800–2,400x reduction
Eviction decision latency	200–600 μs (SW syscall)	~50 ns (HEPC MMIO)	4,000–12,000x reduction
Prefix KV memory @ C=10 requests	C x prefix_KV_GB	1 x prefix_KV_GB	10x reduction (C-fold)
GPU pipeline stalls (KV miss)	1 per T1 miss (200–600 μs)	~0 (async NMCE + prefetch)	Effectively eliminated